

Registers - 32 registers with each being 32/64 bits

a: b → L1 cache

↓ L2 cache

↓ L3 cache

↓ RAM

↓ hard disk

speed decreases
cost decreases
memory increases

→ each layer acts as a cache for the layer below

LRU cache

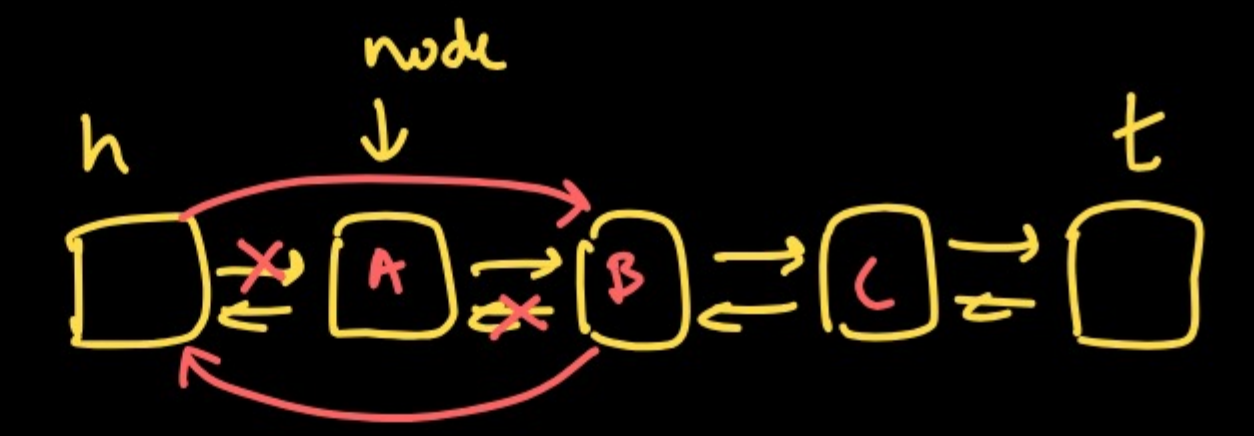
↓
Least recently used

cap: 3

LFU cache

↓
Least frequently used

a: 3
b: 4
c: 1 ← latest



c: LRU cache (2)

c.put(1,1)

c.put(2,2)

c.get(1) // 1

c.put(3,3)

c.get(2) // -1

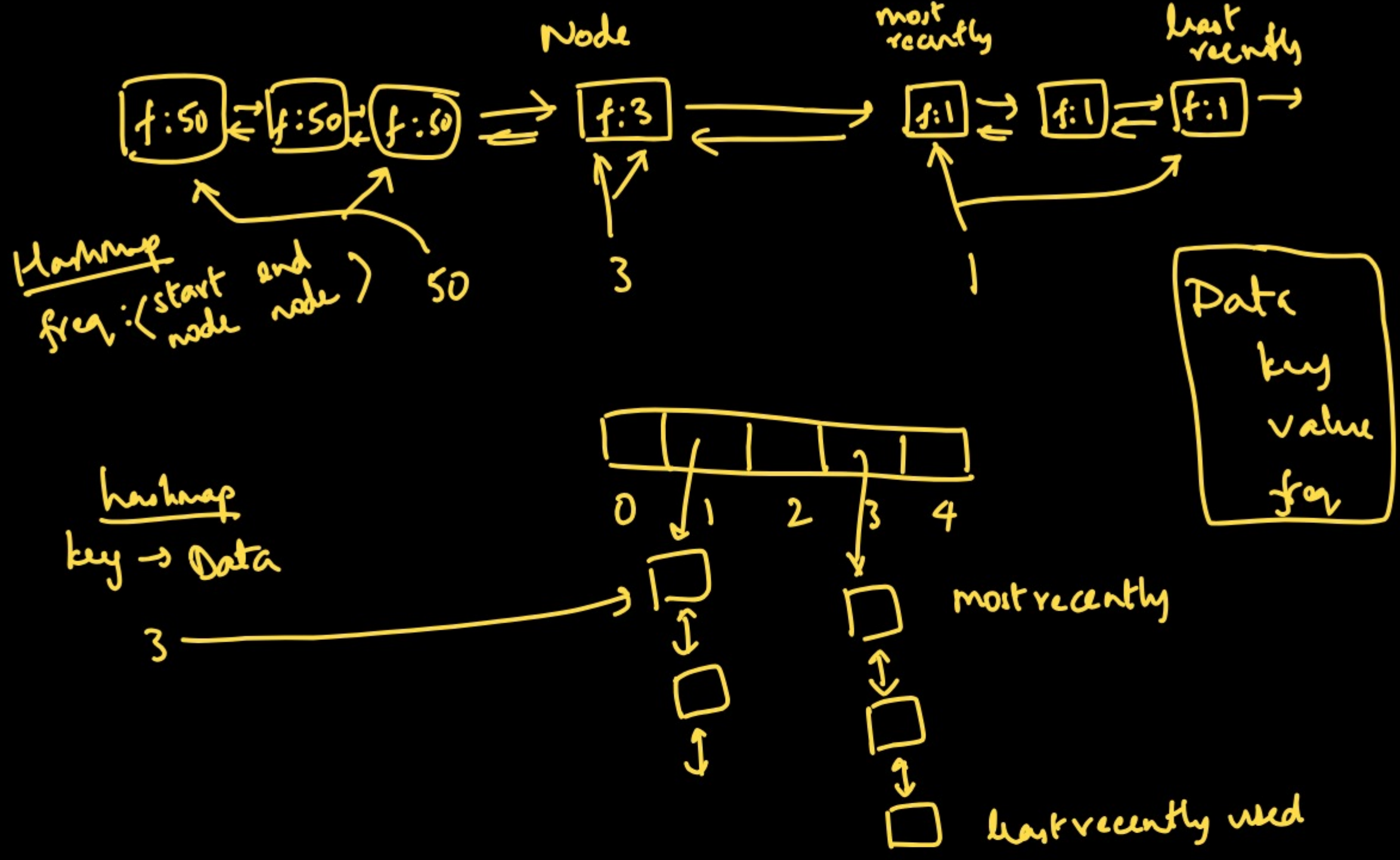
c.put(4,4)

c.get(1) // -1

c.get(3) // 3

c.get(4) // 4

~~1: 1~~
~~2: 2~~
3: 3
4: 4



• HashMap: key → Data
 HashMap: freq → linkedlist
 min Freq Capacity
 addFirst O(1)
 deleteLast O(1)
 deleteNode(Node) O(1)

HashMap < Int, LinkedList
 Pair < Node, Node >

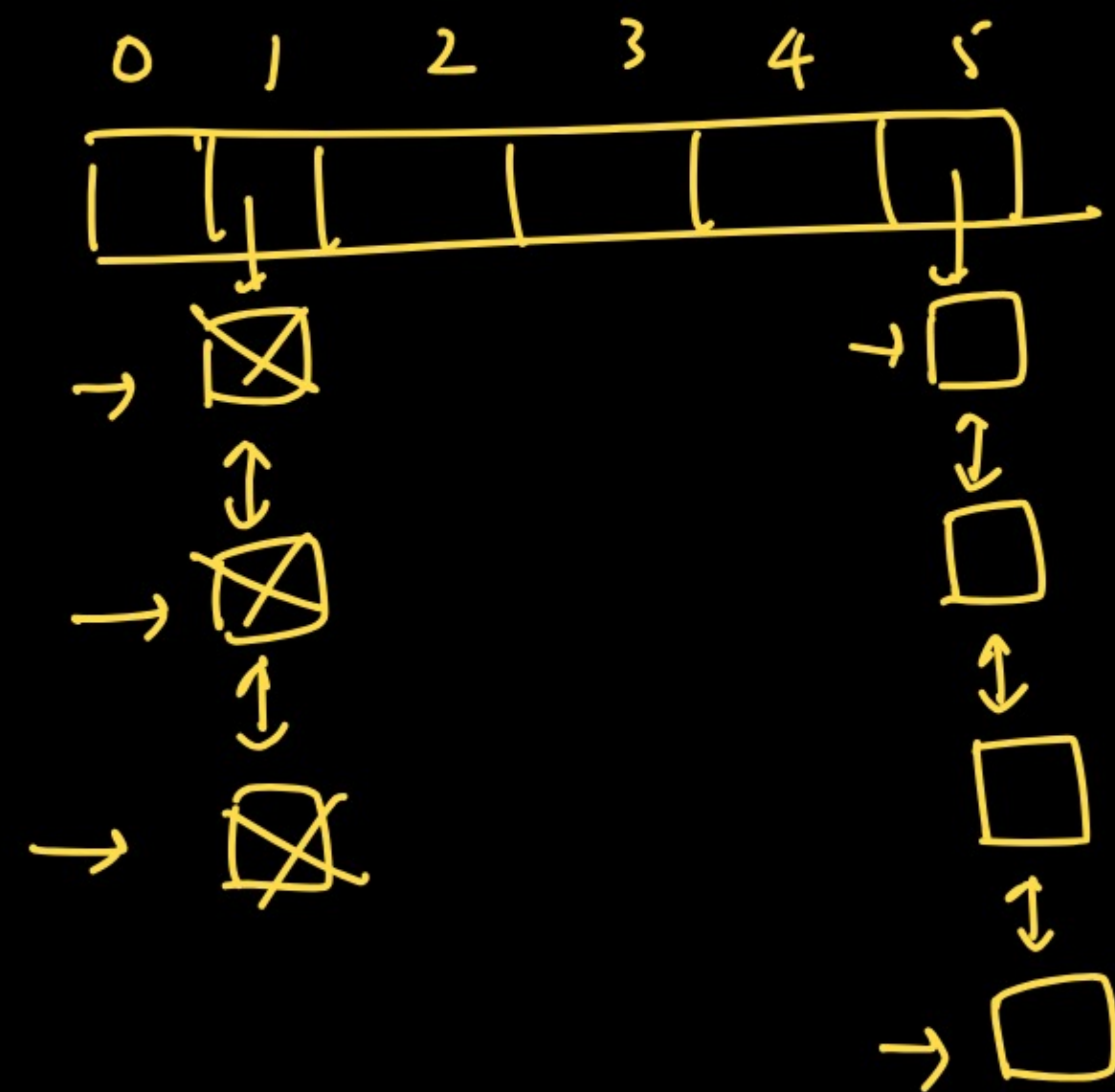
class LinkedList < T > {
 class Node {
 Node data;
 Node next;
 Node prev;
 }
 }

HashMap: from key → Node
 HashMap: from freq → Node
 (head of the linked list)
 Node: tail of lowest frequency

java.util.LinkedList

```
LinkedList<Integer> list1 = new LL<>();
```

LinkedList<Node>
↳ Node



r

	c								
	0	1	2	3	4	5	6	7	8
0									
1		0			1			2	
2									
3		3			4			5	
4									
5									
6									
7		6			7			8	
8									

r=5, c=7
boxIndex=5
rs=1
cs=2

rs=5, cs=4
boxIndex=4
rowSector=1
colSector=1

sector = (rowSector * 3) + colSector

(r, c) → rs = ⌊r/3⌋ cs = ⌊c/3⌋
si = (rs * 3) + cs

1 row ⇒ {'1', '2', '6', '7'}

boolean[10]

-	T	T				T			T
0	1	2	3	4	5	6	7	8	9

nr=6

	nc=4			
	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11
3	12	13	14	15
4	16	17	18	19
5	20	21	22	23

r=2
c=2 ↔ 10

index
r=3
c=1 ↔ 13

index = (r * nc) + c

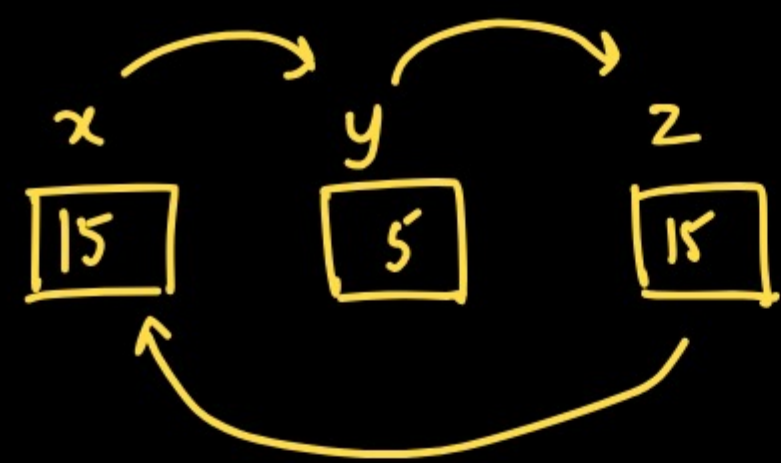
r = ⌊index / nc⌋
c = index % nc

0	1	2	3	4	5	6

element's index = 4
⇒ #elements before it = 4

	cs=2		
rs=1			
	6	7	8

r = 3 * rs; r < 3 * rs + 3; r++ ⇒ r goes from 3 to 5
c = 3 * cs; c < 3 * cs + 3; c++ ⇒ c goes from 6 to 8



temp 10

temp = x
x = z
z = y
y = temp

temp = y
y = x
x = z
z = temp

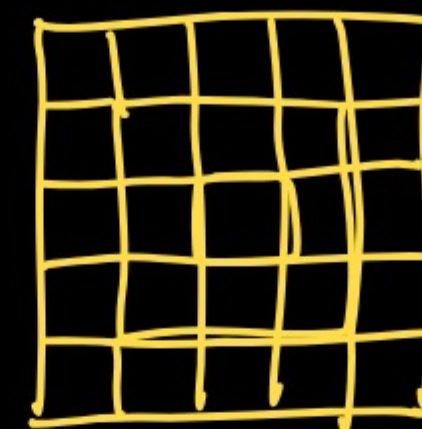
s = 1
↓
(s, s)



m = n = 6

first square $\Rightarrow s = 0$
second $\Rightarrow s = 1$
third $\Rightarrow s = 2$

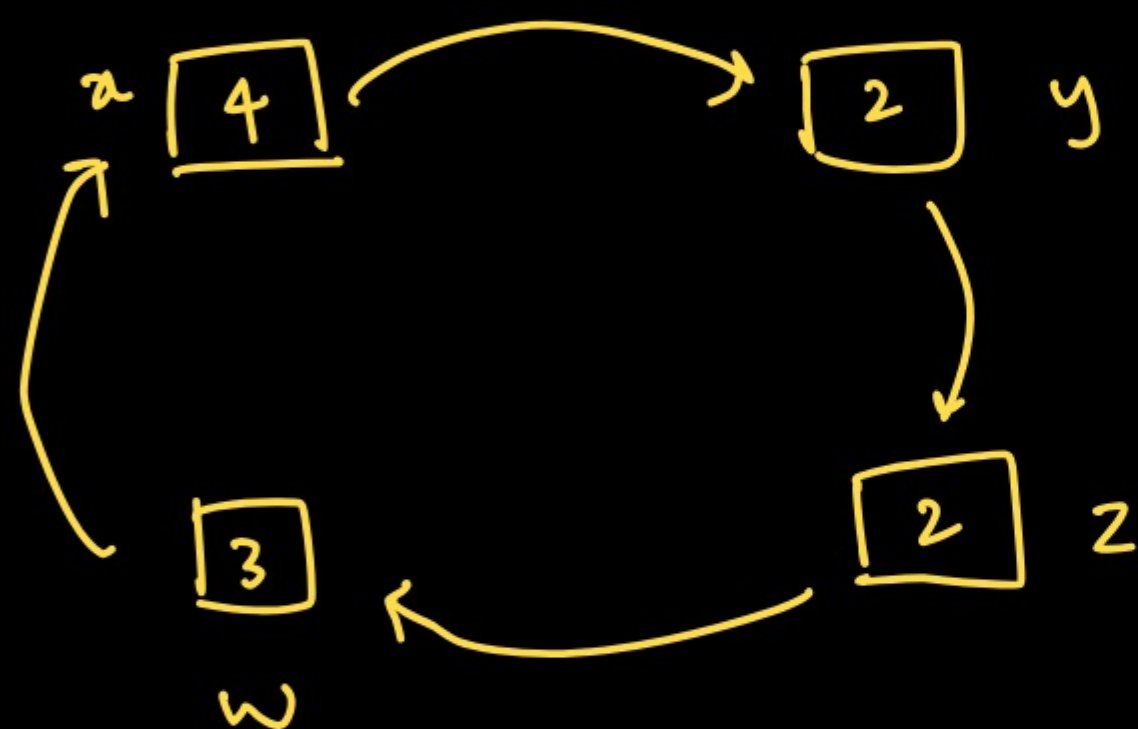
$s \rightarrow 0 \text{ to } \frac{n}{2} - 1$
0 to 2



m = n = 5

s = 0 $\Rightarrow$ first
s = 1 $\Rightarrow$ second

0 to $\frac{5}{2} - 1$
0 to 1

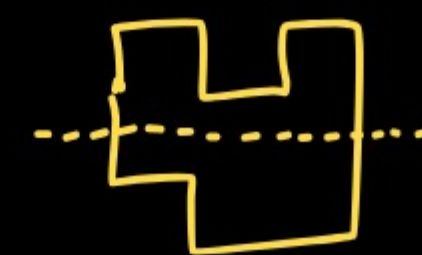
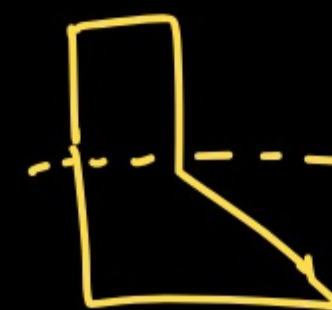
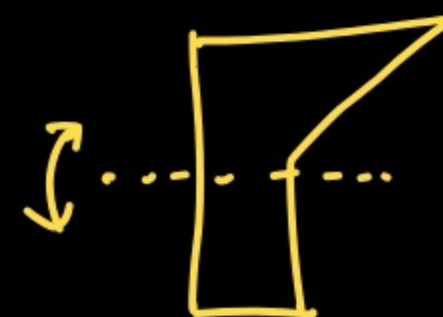
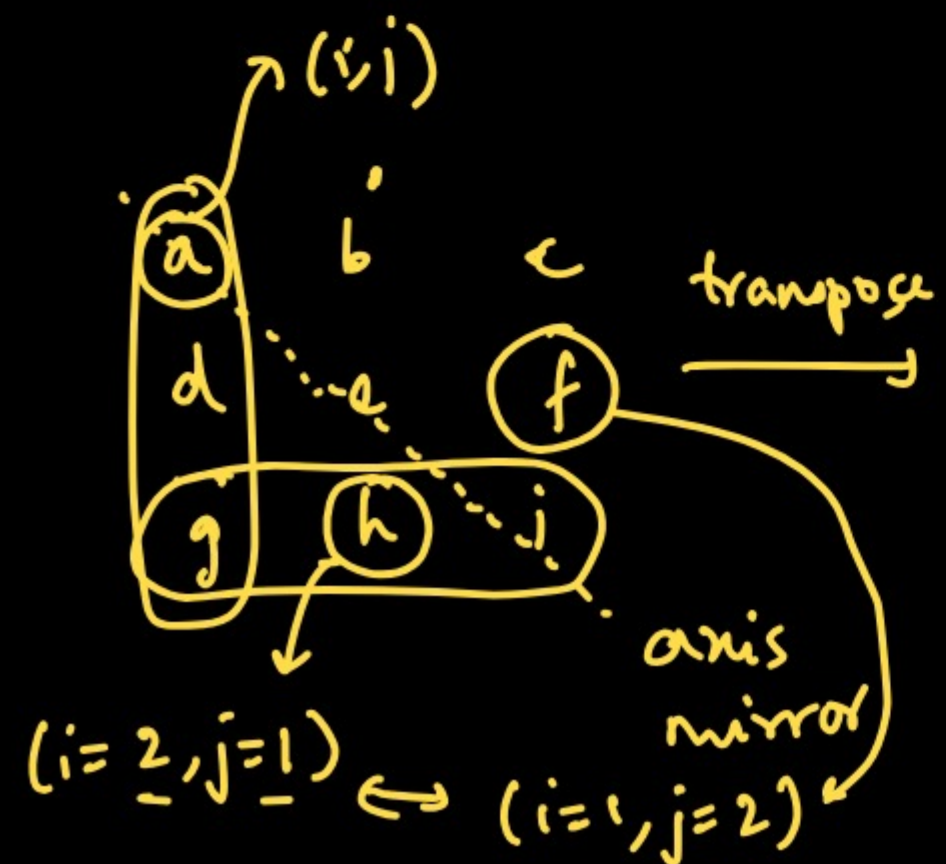


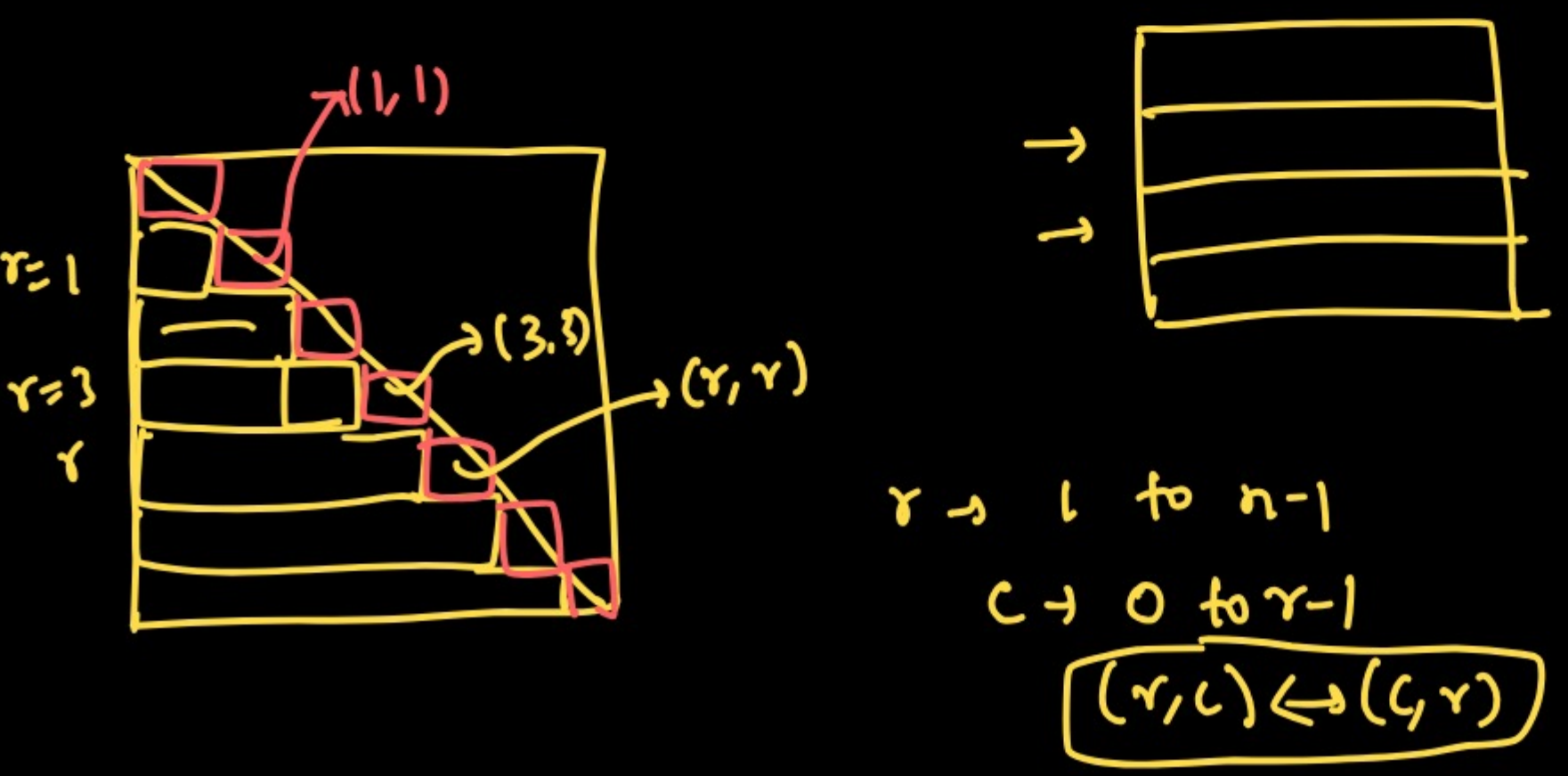
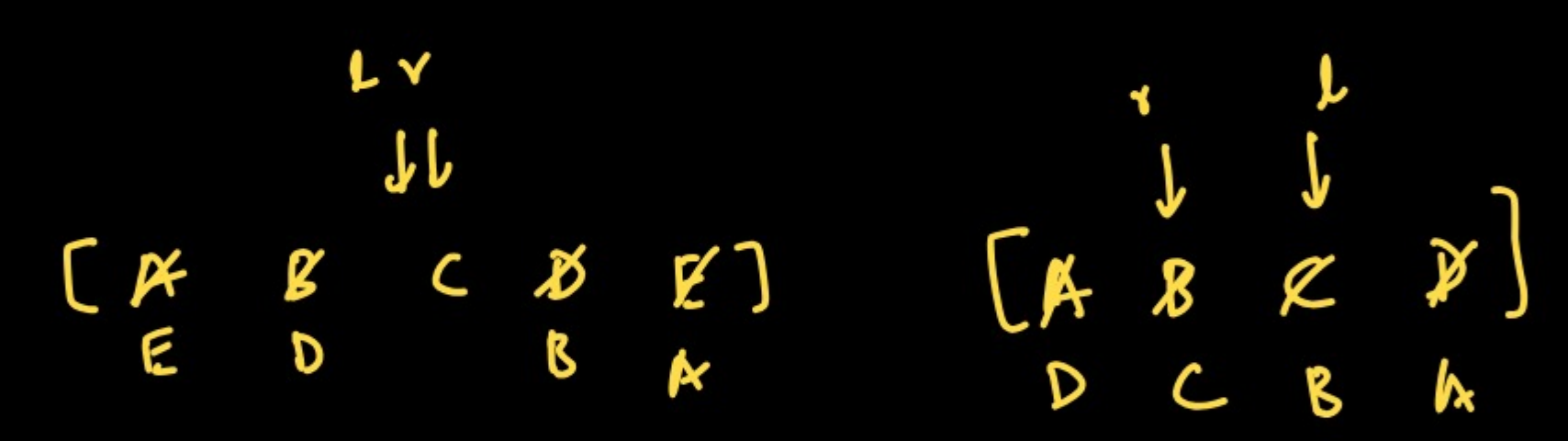
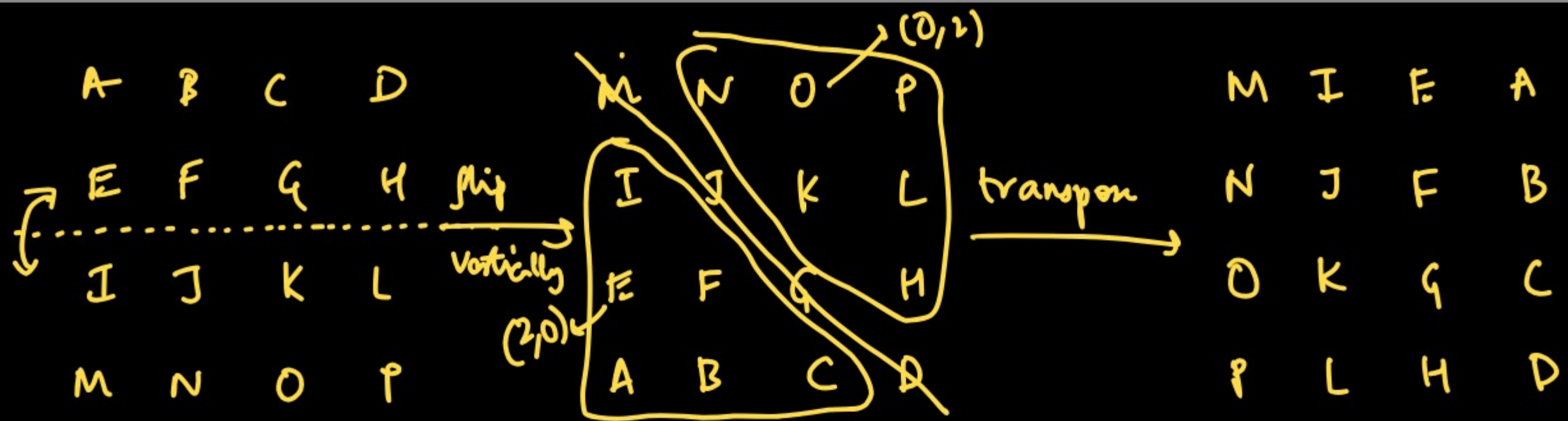
temp 1 1 variable

temp = x
x = w
w = z
z = y
y = temp
5 steps

A B C
... D E F ...
G H I
flip vertically

G H I
D E F
A B C
transpose
G D A
H E B
I F C





$r \rightarrow 1 \text{ to } n-1$
 $c \rightarrow 0 \text{ to } r-1$
 $(r,c) \leftrightarrow (c,r)$

